



Information Asymmetry

When one side of a deal knows more than the other — and what that does to price, risk, and trust.

What is information asymmetry?

“Information asymmetry is when one side of a deal knows something important — about quality, risk, or intent — that the other side doesn’t.”

A simple example: you’re buying a used car. The seller has driven it every day and knows exactly what’s wrong with it. You get a 20-minute test drive. That gap in knowledge is information asymmetry — and it changes what price you’re willing to pay.

Why it matters: whenever it’s costly to verify something, the informed side has an edge — and markets, prices, and contracts all adjust around that gap.

Two types of information asymmetry

The gap shows up at two different moments in a deal — we'll take each one in turn:

BEFORE THE DEAL

Adverse Selection

The informed side already knows something before signing — a seller knows the car's a lemon, an applicant knows their own health risk. The riskier side is often the one most likely to show up.

→ Slides 4–6

AFTER THE DEAL

Moral Hazard

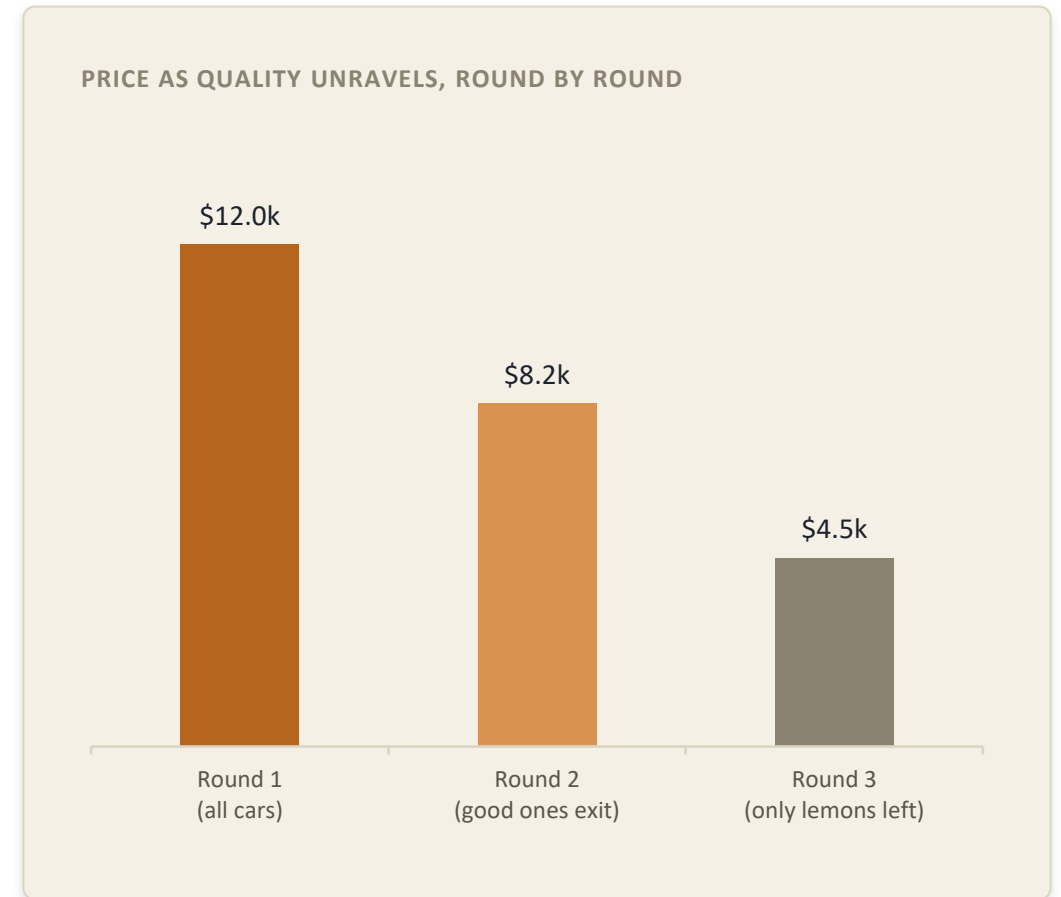
The informed side's behavior changes once the deal is signed — an insured driver takes more risks; a manager relaxes once no one's watching closely.

→ Slides 7–9

The classic “lemons” problem

Adverse selection happens **before** a deal is made — the side with more information already knows something the other side doesn’t. The used-car market is the classic case: sellers know a car’s true quality, but buyers can’t tell a reliable “**peach**” from a defective “**lemon.**”

- 1 Buyers can’t verify quality, so they’ll only pay an average price for the whole pool of cars.
- 2 Owners of the best cars won’t sell at that price, so they walk away.
- 3 That drags the average quality — and price — down even further, and the cycle repeats.



Not just used cars

1

Health insurance

People who expect big medical bills buy the most coverage, pushing premiums up and healthy people out.

2

Credit markets

The riskiest borrowers are often the most eager to accept a high interest rate.

3

Hiring

Without good screening, weaker candidates — who know their own limits best — tend to apply more aggressively.

4

Insurance death spiral

As healthy customers leave a pool, prices rise again, pushing out the next-healthiest group too.

5

IPO underpricing

Founders know more about their company than new investors, so shares are often priced a bit low to attract buyers.

6

Peer-to-peer lending

Early online lenders struggled because the riskiest borrowers were also the most eager applicants.

When lemons take over the market

When buyers can't trust average quality, the best sellers exit first — and if nothing intervenes, the whole market can shrink toward nothing.



Prices collapse toward the worst case — As shown two slides back, prices spiral down until only the lowest-quality option is left to sell.



Good trades never happen — Buyers and sellers who'd both benefit from a fair deal never find each other, because trust breaks down first.



Whole markets can disappear — In the extreme, adverse selection can shut a market down completely — no listings, no buyers, no trade.

How it's usually fixed: screening, signaling, and third-party certification — covered later in this deck.

When the deal itself changes behavior

Moral hazard happens **after** a deal is made. Once part of the cost or risk shifts to someone else, people quietly behave differently — usually taking on a bit more risk than they would on their own dime.

BEFORE THE DEAL

Full risk, full caution

You drive carefully, because any accident is entirely your own cost to bear.

AFTER THE DEAL

Shared risk, less caution

Once you're insured, a fender-bender is mostly the insurer's cost — so a little less caution quietly creeps in.

Beyond car insurance

1

Insurance

Once you're covered, it's tempting to take a few more risks — you're no longer paying the full cost yourself.

2

"Too big to fail" banking

Banks that expect a bailout take on riskier bets than they would if they alone bore the losses.

3

Employment

Without close oversight, people may put in less effort once their job starts to feel secure.

4

Co-signed loans

A guarantee can make a borrower less careful, since someone else is on the hook if things go wrong.

5

Rented & leased assets

A renter has less reason to take care of something than an owner who bears the full cost of wear and tear.

6

Limited liability

A capped downside can push a company toward bigger bets, since the worst losses land on others.

When risk quietly piles up

When someone else bears the cost of your risk, it's rational to take on more of it — and across a whole system, that adds up.



Costs rise for everyone — A few people taking on more risk pushes up premiums, fees, or rates for the whole pool, not just the risk-takers.



Risk builds up unseen — Because the behavior change is often invisible until something goes wrong, risk can quietly accumulate for years.



Small problems can turn systemic — At a large enough scale, this same dynamic helped trigger the 2008 financial crisis — covered next.

How it's usually fixed: monitoring, deductibles and co-pays, and contracts that tie pay to outcomes — covered later in this deck.

The 2008 mortgage crisis

1

Mortgage originators sold loans onward almost immediately, so they bore little cost if a borrower later defaulted.

2

Rating agencies were paid by the very issuers whose securities they rated — inflating many AAA ratings.

3

Investors, unable to verify true loan quality themselves, relied on those ratings as their main signal of safety.

4

When default rates rose, trust in the ratings collapsed — and demand for these securities unraveled almost overnight.

Both types at once: moral hazard at origination (lenders bore little of the downside), and adverse selection downstream (buyers of the bundled securities couldn't tell good loans from bad).

How businesses close the gap

SIGNALING

Prove it, don't just claim it

The informed side takes a costly action that only a genuinely good option could afford — proving quality instead of just claiming it.

Examples: a college degree, a money-back guarantee, a company's brand reputation.

SCREENING

Let people sort themselves

The uninformed side offers a set of choices designed so different people naturally pick the option that fits them.

Examples: insurance plans with different deductibles, airline fare classes, freemium software tiers.

Reputation — Repeat players have more to lose by cheating.

Certification — Ratings, warranties, and inspections verify what you can't.

Regulation & monitoring — Disclosure rules and oversight put the cost of hidden problems back where it belongs.

Why this matters

Online marketplaces

Star ratings substitute for the quality-checking buyers can't do themselves.

Health insurance

Deductibles and tiers manage both types of information asymmetry at once.

Startup fundraising

Founders know their venture's prospects far better than investors ever can.

Everyday negotiation

If you can't verify what the other side knows, assume they've priced it in.

In one line:

“Markets rarely fail from asymmetric information — they adapt to it, through signals, screens, and reputations that make the invisible costly to fake.”

Where this all began

“The Market for ‘Lemons’: Quality Uncertainty and the Market Mechanism”

George A. Akerlof · Quarterly Journal of Economics, 1970

This is the paper that started it all. Akerlof used the used-car market to show, formally, how quality uncertainty alone — with no bad intentions on anyone’s part — can drag prices down and push good sellers out, sometimes collapsing a market entirely. It was turned down by three journals before the QJE accepted it, partly because referees felt a “used cars” model was too trivial for economics.

2001 NOBEL PRIZE

Awarded the Nobel Memorial Prize in Economic Sciences, shared with **Michael Spence** (signaling) and **Joseph Stiglitz** (screening), for laying the foundations of information economics — the field this whole deck comes from.